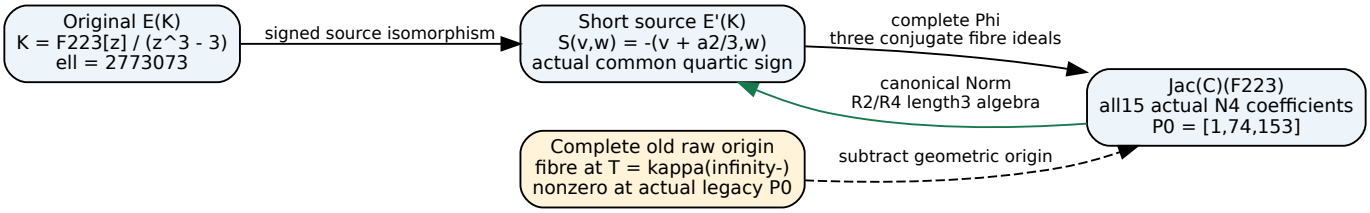
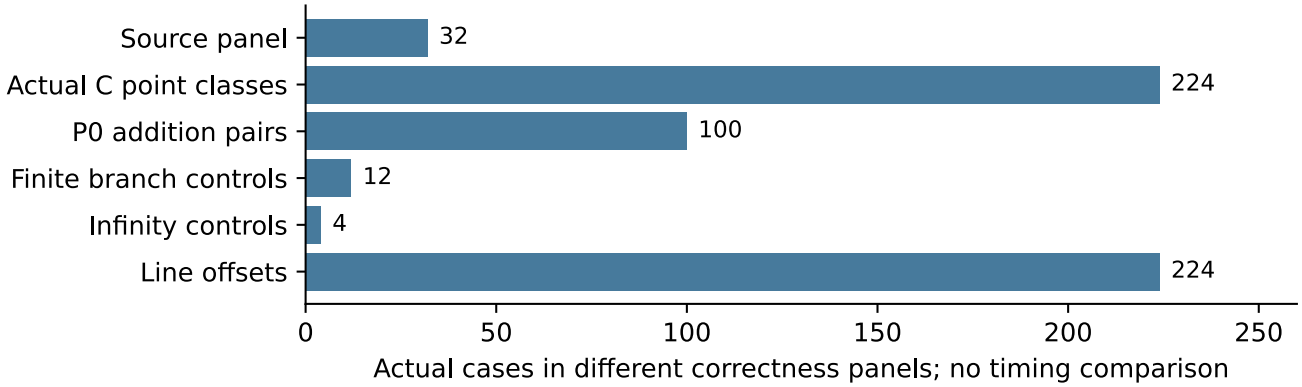
Actual q223 completed maps

2026-11-13 · bounded native evidence · all standing unsigned





Observed unsigned a5a4a6: one comprehensive unfiltered Rust test passes, zero failed/ignored/filtered, all2203 input guards unchanged. The additive V2 constructors rebuild the actual q223 field, plane quartic, P0, normal forms and complete origin. Original V1 and all q5/7/11 controls remain unchanged. Four original N4 files and the frozen P1 algorithm remain byte exact.

Exact source: K=F223[z]/(z³−3), ell2773073, #E(K)=11092292=4ell. All basis products/Frobenius and all15 normalized quartic coefficients match the actual N4 source. Canonical P0=[1,0,0]; the original winning FB at seed102/nfb36 actually chooses [1,74,153]. Both completed constructors use their own true P0 and model tags. Source j is outside F223; this is a restricted extension-field family.

The entire actual four-element 2-Sylow subgroup is checked, including all three original cubic two-torsion roots. Eight odd multiples crossed with it give32 source controls. Both [2] compositions hold on the stated source/class panels, and Phi(G) is nonzero with exact prime order2773073. These are bounded native checks, not enumeration of the11092292 source or Jacobian elements.

Origins, proofs and independent checks

Actual sign/origin: completed kappa and legacy lambda use opposite quartic ordinates. I(v,w)=(v+a2/3,w), T=kappa(negative infinity), and kappa(lambda⁻¹P)=T−I(P). After subtracting the actual old infinity fibre, Phi_old=Phi_new∘(-I). Here T=((116,126,164),(133,115,44));2T is nonzero. All16 admitted split fibres in the panel equal completed raw fibres at each actual P0. At the winning legacy P0 the raw origin is nonzero; it is constructed geometrically, never fitted.

Reference divisors: Dinf=K_C−P0. A fixed final divisor key shifts by delta=[Pnew−Pold], while the same raw degree-six space shifts by2delta. Native controls check zero/inverse,100 addition pairs, the raw-degree6 formula and the complete map bridge at distinct actual P0. Fresh norm at the legacy P0 checks32 source points, all five public point half-inverses and all five affine three-point cancellations; the ratio base is nonzero.

Exceptional controls: all224 plane-line offsets agree, all12 geometric roots from the three quadratics and all four infinity signs pass. The actual3P scheme at[1,1,52] retains multiplicity3; discarding multiplicity changes its norm, and an alternate affine chart agrees with the correct result. The first3P at[1,0,0] is a retained null control: omission gives the same norm there. No claim turns that null into a discriminating negative control.

Written proof and formal boundary: pi_i* pi_i^*=[2]; cross terms vanish by the other Ciani involutions. The sum of three pull-push projectors is2 plus the V4 norm, which factors through Pic^0(P1)=0. Hence both compositions are[2]; on the odd subgroup, inverse2=1386537 recovers points and preserves DLP scalars. V2_MAPS_01.md spells out the actual divisor/model origins. Existing9+5 Lean bindings prove conditional module/ring identities, not the actual Picard, Hasse, cover or N4 solver objects. Full formal geometry remains open.

Independent controls now settled at their scopes: full replay67baa2 passes all43 phase pairs (29positive0/0+14intentional2/0), both original9+5 proof gates,14 exact bindings and435 guards. Exact68d ownCI43432174569/job131908108813 succeeds;13 native records match after excluding only recorded wall/wall_total values,14 canonical claim digests match (a5a143). Rho accessor e5d852 checks all five recovered native/actualCLI answers; the walk is unchanged by its three output-only substitutions.

Failures and next action: whole full01 consumer-path failure080841 and rho01 counter-boundary/timestamp failure638d96 remain, with corrected passes and all original sources. Seven complete closed captures and seven full member indices are retained; genuine remote publication is separately evidenced. The active prospective protocol74e00a/COMPLETED_DESCENT_PROTOCOL_02.md replaces only split-transfer triple admission in individual descent, after normal publication. Map overhead may outweigh avoided rejection; original counters omit arithmetic in the separate completed-map engine. No benchmark, computational advantage, generic prime-field coverage or signed execution is established.

Source-linked evidence:

a5a4a6/V2 native02;67baa2/full02;a5a143/ownCI;e5d852/rho02;7da26d/extension freeze;V2_MAPS_01.md;existing f68241+d71088 partial formal bindings. Whole failed and passed captures/member indices remain separate immutable evidence.